Unit 4, Lesson 9: Finding the Area of a Circle

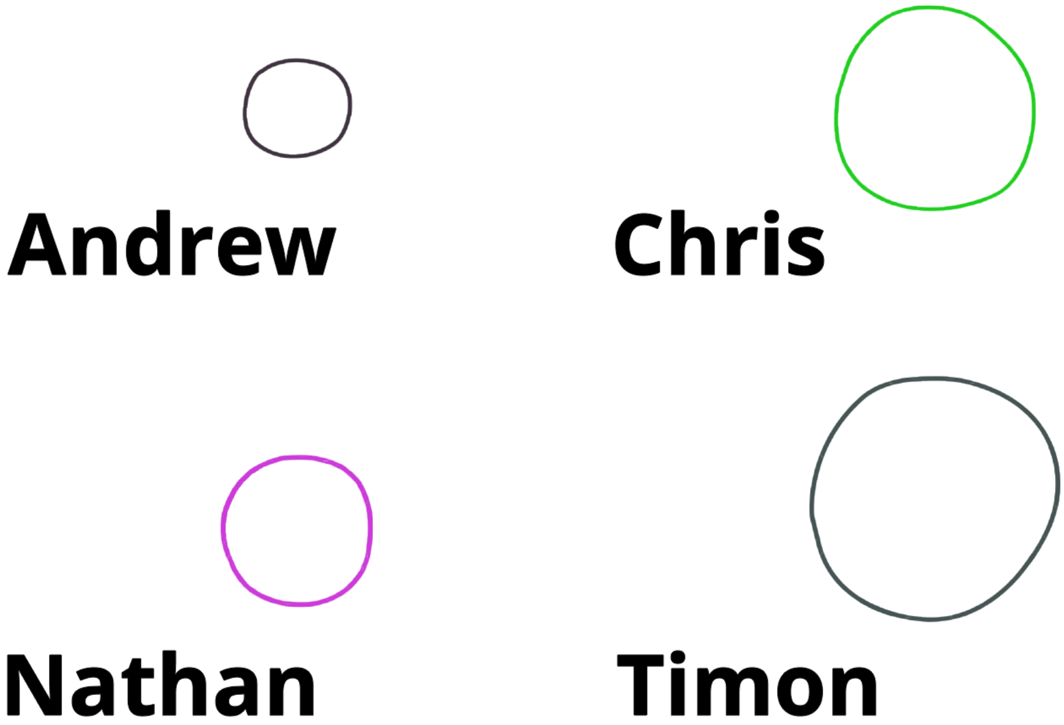
Benchmark

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

Learning Target

☐ I can find the area of a circle given the radius or diameter.

4.9.1 Warm-Up: Which Is the Best?

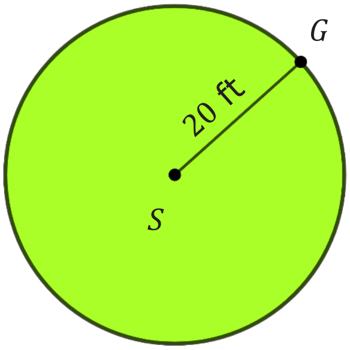


1. Rank each person's image based on which looks most like a circle.

2. Explain how you decided upon your ranking.

4.9.2 Guided Instruction: Finding the Area of a Circle

1. A goat, located at point G, is tied with a 20 foot (ft) rope to a stake located at point S.



a. Complete each statement with words from the word bank.

$A = \pi r^2$	radius	$C = 2\pi r$	area	circumference

The space in which the goat can roam is the ____________ of the circle. The distance of the rope from the stake to the goat is the ____________ of the circle. To find the total space in which the goat can roam, use the formula ____________.

b. What is the area over which the goat can roam?

The **area of a circle**, A, can be found using the formula $A = \pi r^2$ or $A = \frac{1}{2}Cr$, where C is the circumference of the circle, π is the proportional relationship between the diameter and the circumference of any circle, and r is the radius.

2. Jeffery, Katelyn, and Azia each found the area, in square feet (ft^2), in which the goat can roam. Their work is shown.

Jeffery's Work	Katelyn's Work	Azia's Work
$A = \pi r^2$ $A = \pi(20^2)$ $A = 400\pi$ ft^2	$A = \frac{1}{2}Cr$ $A = \frac{1}{2}(2\pi \cdot 20) \cdot 20$ $A = \frac{1}{2}(2 \cdot 3.14 \cdot 20)(20)$ $A \approx 1256$ ft^2	$A = \pi r^2$ $A = \pi(20^2)$ $A = \left(\frac{22}{7}\right)(400)$ $A \approx 1257.14$ ft^2

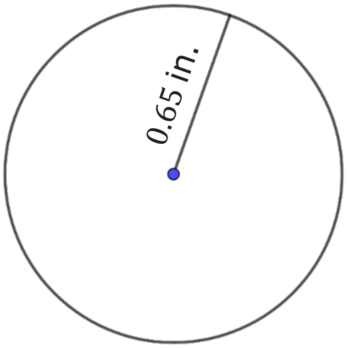
a. Explain why Jeffery uses the equal sign (=) but Katelyn and Azia both use the approximate sign (≈).

b. Harold said that he used the π button on his calculator instead of using an approximate value for π. What answer did Harold get, rounded to the hundredths place?

When finding the area of a circle, always write answers using the representation of π based on the directions given in the problem.

4.9.3 Guided Practice: Finding the Area of a Circle

1. A circle with radius 0.65 inches (in.) is shown. Complete the table by finding the area of the circle using the given value for π.



Approximation of π	Area
3.14	
$\frac{22}{7}$	
π button on the calculator	

2. Roberto wants to replace the floor on his circular skating rink but doesn't know the area. He knows the length across the skating rink through the center is 60 meters (m). He asked his two sons, Adrian and Juan, to help him determine the area. Their work, rounded to the nearest hundredth, is shown.

Adrian's Method	Juan's Method
$A = \pi \cdot 60^2$ $A = \pi \cdot 3600$ $A = 11{,}309.73$ m^2	$A = \frac{1}{2}(\pi \cdot 60) \cdot 30$ $A = \pi \cdot 900$ $A = 2827.43$ m^2

a. Whose method is correct?

b. Explain the mistake that was made to the son who made the error.

4.9.4 Your Turn!

1. Elena wants to tile the top of a circular table. The radius of the top is $14\frac{5}{8}$ inches (in.). What is the area of the table that will be tiled? Use $\frac{22}{7}$ for π.

2. Jada painted the face of a circular clock that has a diameter of $2\frac{4}{5}$ feet (ft). What area of the clock was painted? Use 3.14 for π.

3. A circle has a diameter that measures $7\frac{9}{10}$ centimeters (cm) in length. What is the area of the circle? Use the π on your calculator.

4.9.5 Wrap-Up: Error Analysis

1. Clare and Andre are baking and frosting cookies.

Clare says:	Andre says:
"The radius of my cookie is about 3 cm, so the space for frosting is about 18.8 square centimeters (cm^2)."	"The diameter of my cookie is about 3 in., so the space for frosting is about 28.3 square inches (in^2)."

a. Who, if anyone, correctly approximated how much space there is to spread frosting on their cookie?

b. Write to either Clare or Andre an explanation of why their statement is incorrect.